MANAV PREET SINGH

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SKILLS

Languages: Python, SQL, Java, JavaScript, TypeScript, Bash
Data & Analysis: Pandas, NumPy, Scikit-learn, data validation, anomaly detection, EDA, Grafana dashboards, reporting
AI / ML: LLMs (OpenAI, Groq, self-hosted WizardLM, Ollama / Qwen2.5), RAG, LangChain, agentic workflows, prompt engineering, PyTorch, Hugging Face
Cloud & Databases: PostgreSQL, MySQL, MongoDB, AWS (S3), GCP (Cloud Run, GKE, Cloud Build), Docker, CI/CD, GitHub Actions, Kubernetes (basic)
Other: REST APIs, Git, Agile/Scrum, Grafana, Linux

EXPERIENCE

J.D. Power May 2026 – Present
Software / QA Automation Engineer *London, ON*

- Migrated CI/CD pipelines from templates to reusable components, standardizing build and test stages into versioned, shareable building blocks.
- Validated an internal Copilot-powered tool that generates full software test cycles from inputs such as Jira tickets and freeform text, assessing the accuracy and coverage of AI-generated test cases.

J.D. Power May 2024 – Aug 2025
QA Automation Engineer Intern (Software Engineering / ML / Data scope) *London, ON*

- Developed Python ML services using Random Forest and statistical methods to flag outliers in vehicle pricing and logic-tree datasets, scanning thousands of daily records to surface data quality issues before they reached downstream systems, directly analogous to data validation on AI-generated outputs.
- Built Java backend metrics pipelines with Grafana dashboards consolidating production database activity into an analytics layer, enabling stakeholder visibility into Copilot usage, git activity, and engineering productivity across QA and software teams.
- Automated daily AWS S3 backups of Copilot metrics datasets, building a longitudinal usage archive that enabled trend analysis and protected against data loss.
- Extended internal QA tooling with REST API integrations for Jira sync and bulk import/export, replacing manual data entry with automated pipelines and keeping datasets in sync across engineering teams.
- Presented end-to-end ML anomaly detection findings to VP and senior leadership, walking through model architecture and detection patterns on live production data to drive adoption of the initiative.

Lawson Health Research Institute Sept 2023 – Apr 2024
Data Analyst (NLP / ML) *London, ON*

- Designed a RAG pipeline that retrieved only the most relevant clinical text segments (medications, diagnoses, vitals) before LLM extraction, reducing hallucinations and improving data accuracy across variable, unstructured clinical note formats.
- Built a privacy-preserving Python data extraction system using self-hosted LLMs (WizardLM), converting unstructured clinical narratives into structured, queryable records, a direct application of GenAI for data transformation and validation.
- Used LangChain, custom prompts, few-shot examples, and parsing logic to implement an extraction pipeline that reduced clinical note review time by ~50%, clearing a research record backlog and freeing analyst time for higher-order work.
- Validated model outputs against ground-truth records and iteratively refined prompts based on exploratory analysis of failure patterns and edge cases across diverse note formats.

PROJECTS

LEVEL UP | AI Career Prep Platform (Software Engineering Capstone) Sept 2025 – Mar 2026
Capstone grade: 95 / Dean's Honor List · Team of 4, advised by Dr. Alexandra L'Heureux

- Built a full-stack web platform with React, FastAPI, and PostgreSQL, integrating Groq and Hugging Face APIs to deliver resume tailoring, RAG-grounded cover letters, and interview practice in a single workflow.
- Designed LLM workflows to extract job-description keywords, map them to resume evidence, and score candidate submissions, implementing an LLM-as-judge evaluation pipeline to assess and improve output quality.

EVENTOPS AI | Agentic Hackathon Planning System (SE4471) Fall 2025

- Implemented a RAG pipeline over historical event data with LangChain memory for cross-step coherence; benchmarked self-hosted Qwen2.5-7B against API-hosted models for quality/cost trade-offs.
- Built a React + FastAPI frontend/backend split with streaming responses, enabling real-time visibility into multi-step AI-generated planning outputs.

EDUCATION

University of Western Ontario London, ON
Bachelor of Engineering Science, Software Engineering (Co-op) | GPA: 3.7

Honors: Dean's Honor List (2021, 2022, 2023, 2025) | Heaslip Scholarship (2022, 2023, 2025)

Selected Coursework: Database Management, Machine Learning, Deep Learning & Computer Vision, Cloud Computing, Software Testing & Maintenance, Algorithms & Data Structures, Information Security, Software Design